YMLS 2025 fall: Deep Learning and RL Exam Procedure

Exam procedure

- Our exam procedure looks like a small technical interview for an ML Engineer position.
- Exam duration: 30-40 minutes, 4-5 minutes for preparation for each question (you also may start answering right away, if you want).
- You will get 1 question from the first half of the exam set and 1 question from the second half of the questions set.
- Once you know your exam ticket (set of 2 questions) you will have 5 minutes to self-prepare or can start answering right away. The total maximum points you can get for this part is **70 pts.**
- You may be asked beyond your ticket questions for adjacent topics, based on the decision of the examiner. E.g., if your ticket is about Linear Regression, you can get small extra questions on Logistic Regression or Metrics, Naive Bayes.
- During the answer, you need to follow the sequential logic of your narration and give all the formulas, definitions and concepts/theorems you use. The deeper your answer is, the more points you will get.
- You will have paper sheets to write down your answers, please bring your own pen.
- If you want to get 70pts.+ or want to increase your grade for the exam, you may continue with 1 extra question taken randomly from the questions set (except a couple of questions you already answered). You can get extra **30 pts.** for this part. This action may be done only after the examiner's permission – please ask him in advance, and/or he can decide proactively that you need to pass 3rd part on his own. Please, be informed, that you also may <u>decrease your points</u> if you will not be able to properly answer the 3rd question – so please, take this option wisely.
- You will have an exam passing schedule in advance, so you need to be prepared and need to be at the university at least **20 minutes before** your exam timeslot. Otherwise, you may not be allowed to pass the exam.

Complete formula to calculate your final grade for the ML course:

$$\boxed{\mathrm{Final} = \min\left(0.5 \cdot \mathrm{Semester} + 0.5 \cdot \mathrm{Exam}, 100\right)}$$

Important notice: some of the students may get more than 100 for *Semester* or *Exam* components for their excellent results during the semester or exam, as well as might be penalized if the result on the exam will be significantly bad.

Passing criteria: The final grade will still be not greater than 100, refer to the formula above. Passing score for the course is 50 Final pts., at least 3 any homeworks done from 6 (4 HWs and 2 Labs) AND passed the exam with at least 30 pts.

Appeals procedure: Right after the exam you have an option to appeal your results if you are not satisfied with them or have an opinion that the exam process was not fair. You need to report to your Academic Head and describe your appeal case. All the appeal cases will be resolved individually, but please take it wisely as well: your points might be decreased after the appeal procedure.

Worst case scenario: If you will not be able to pass the exam (got less than 30 final pts. during the exam) you will be asked to answer some questions (2-5) from the theoretical minimum questions set (see below at the end of this document). You may get up to 5-10 pts. for each question, but no more than 30pts. total for the exam.

Cheating: **If you will be caught cheating or on plagiarism/talking with your groupmates during the exam, you will be deleted right away without the option to appeal or retake the exam.**

Retake procedure: **There is NO retake procedure provisioned for this subject.**

Exam questions list

Natural Language Processing (1st question)

1. Text feature extraction: Bag of Words, Bag of Ngrams, Tf-Idf. Normalization techniques.
2. Word embeddings. Word2Vec. CBOW, Skip-Gram.
3. Word embeddings: Matrix factorization, PPMI.
4. Metric learning, SimCLR. Semantic search.
5. Sequence-to-sequence models. Seq2seq training. Decoding strategies (beam search, greedy, sampling - *on a high-level, just understanding of these techniques).

6**. Attention in seq2seq.

7**. Machine translation task. MT metrics, training pipeline.

8**. Transformer architecture. Self-attention. Transformer block structure. Positional encoding.

9. Transfer learning idea. Contextual pretrained representations (CoVe, ELMo).
10. Transfer learning with pretrained models. GPT-1, BERT .
11. Generative pretraining. T5, GPT-2/3/4. Zero-shot and few-shot inference. Prompting strategies (chain-of-thought, self-consistency).
12***. LLM Alignment. Instruction tuning, RLHF .
13***. PEFT: Sparse methods. Pruning, sparse fine-tuning.
14***. PEFT: loRA, prompt-tuning, multilayer prompt-tuning.

Reinforcement learning (2nd question)

1. RL problem statement. MDP formalism. Crossentropy method.
2. Model-based RL. Bellman equations. Policy iteration with dynamic programming.
3. Value-based RL. Model-free prediction (Monte-Carlo vs. Temporal Difference).
4. Value-based RL. Model-free control. Q-Learning, SARSA, EV-SARSA.
5. Approximate value-based methods. DQN.
6***. Policy-based RL. REINFORCE (with loss derivation).
7***. Actor-critic policy gradient. Baselines, Advantage, A2C.
8***. Exploration strategies. Eps-greedy, UCB, Thompson sampling.
9***. Reinforcement learning for seq2seq. Self-critical Sequence Training.

** Questions marked like this are core topics of the course, so students should be able to always answer them - it’s a minimal requirement for all students, applicable even for those, who wants to take Theoretical Minimum

*** You have an option to retake the exam questions if these questions with this mark appear inside them. But taking them and answering them will give **you extra points**.

Theoretical minimum

1. Embedding layer structure. How does backprop through embedding layer work?
2. Word2Vec main idea.
3. Triplet loss, contrastive learning.
4. Metric learning, DSSM main idea.
5. Seq2seq architecture.
6. Autoregressive decoding strategies: greedy, sampling, beam search.
7. Attention mechanism.
8. Transfer learning main idea. BERT pretraining objectives.
9. Generative pretraining, GPT .
10. Why do we need to use human preferences instead of supervised FT in LLM alignment?
11. RL problem statement (environment, agent, state, action). Reward,

discounted reward.

12. *MDP model and its assumptions.*
13. *Model-free and model-based RL difference.*
14. *On-policy and off-policy RL difference.*
15. *Value function and q-function. Relationship between them.*
16. *Exploration / exploitation tradeoff. Describe any exploration tactic.*
17. *Main CV architectures (ResNet, VGG, AlexNet - any of them at least).*